Estimate climate sensitivity

The equilibrium climate sensitivity (ECS) is the longterm response to a doubling of the atmospheric CO2 concentration over pre-industrial levels. Here we shall use short simulations of only 10 years and linear theory, the linearised energy balance equation (Equation 3.4 in course notes), to estimate ECS for the MPI-ESM1.2-CR model. You can choose to apply one or more doublings, e.g. 2, 4 or 8xCO2, whereby the forcing is then 1, 2 or 3 times that of a doubling. You can also try halving the CO2 concentration, in which case you apply minus the forcing from a doubling. Finally, some of you might want to play with the solar constant, whereby a 2 percent increase is roughly the same forcing as a single doubling of CO2.

First, you must conduct a pre-industrial control simulation as described in the guide "Running MPI-ESM1.2 on Tetralith". Then return to the run/ directory and copy the config file you used to create the control simulation to another name, for example:

```
cp examples/piControl-CR.config abrupt8xCO2.config
```

The easiest way to then increase the CO2 volume mixing ratio (`co2vmr`) is to edit the config file and add the following lines:

```
[namelists]
  [[namelist.echam]]
    [[[radctl]]]
      co2vmr = <your_concentration>
```

which tells the the script generator `mkexp` to override the default value in the namelist. To figure out which number to write instead of <your_concentration>, you have to find the default value for the piControl experiment and multiply by whatever factor you have chosen. Remember that these are very small concentrations, so include the 'e-06' after the number. After this use `mkexp` to generate the new experiment and run it.

If you wish instead to change the solar constant you can proceed in a similar way:

```
[namelists]
  [[namelist.echam]]
    [[[radctl]]]
      solc_multiplier = <your_multiplier>
```

where you can insert a multiplier of your liking, e.g. 1.02 for a 2 percent increase in the solar constant.

An alternative to changing the namelist settings of your experiment is to edit the .run and .run_start scripts after you generated them. Then you need to set things equal in both scripts.

As you analyse your results remember to subtract the time mean values from your control simulations, including the radiation balance which isn't zero in the control simulation.

Gregory, J. M. et al. A new method for diagnosing radiative forcing and climate sensitivity. Geophysical Research Letters, 31, L03205, doi:10.1029/2003GL018747, 2004.